ATCD UNIT 3

Context Free Grammars (CFG), Derivations, Parse Trees, Ambiguity

Formal Definition of CFG

(05 Marks)

1. Define context free grammar.
2. Abbreviate and define CFG.

(06 Marks)

1. Define the terms

   (i) Derivation

   (ii) Recursive inference

   (iii) Language of a grammar

   (iv) Sentential form

   (v) Parse tree.
2. Define the following:

   i) Language of CFG

   ii) Sentential form

   iii) Yield of a parse tree.
3. Define Ambiguity.

CFG Construction / Writing Grammar

(05 Marks)

1. Give the formal definition of CFG. Give CFG that generates the strings for the Language L = $\{0^n1^n \mid n >= 1\}$.
2. Construct CFG for accepting the strings containing alphabets a and b, which will start with exactly one 'a' followed by any number of b's.
3. Define CFG. Obtain a CFG for the language

L = $\{a^nb^mc^k \mid n+2m=k, n>=0, m>=0\}$

4 . Design CFG for the language of palindromes over {0,1}.

5. Design CFG over {a,b} consisting of all strings of even length.

(06 Marks)

1. Define CFG for the following:

   (i) Binary strings with twice as many ones as zeros.

   (ii) L = $\{a^nb^m \mid n \neq m\}$
2. Design a CFG to generate:

   (i) the language of the regular expression 0∗(1(0+1).*(ii) L = $\{a^n\omega^RB^n \mid \omega \in \{a,b\}$* }(iii) L = $\{\omega \mid n_a(\omega) = n\_b(\omega)+1\}$
3. Obtain context free grammar on Σ={a,b} to generate a language

   i) L=$\{a^mb^n \mid m \neq n\}$

   ii) L=$\{a^nww^Rb^n \mid w \in \Sigma^*, n\geq1\}$
4. Give CFGs that generate the following languages over {0,1}:

   i) all strings that start and end with the same symbol

   ii) strings of odd length and its middle symbol is 0.
5. Construct CFG for the following language:

   i) L=$\{a^nb^ma^n \mid m,n>=1\}$

   ii) L=$\{a^ib^jc^k \mid i\neq j \text{ or } j\neq k\}$

6. Obtain context free grammar on $\Sigma=\{a,b\}$ to generate a language

   (i) $L=\{a^n b^n \mid m \neq n\}$

   (ii) $L=\{a^n w w^R b^n \mid w \in \Sigma^*, n \geq 1\}$

7. Obtain a grammar to generate the language:

   i) $L=\{a^{n+2} b^m \mid n \geq 0 \text{ and } m > n\}$

   ii) $L=\{a^n b^m \mid n \geq 0, m > n\}$

8. Write Context Free Grammar to specify:

   (i) All strings over (a,b) that are even and odd length and palindromes.

   (ii) $L=\{a^n b^{2n}$ over $\Sigma=\{a,b\}, n \geq 1\}$

   (iii) $L=\{0^m 1^m 2^n;\ m \geq 1, n \geq 0\}$

9. Define Context Free Grammar. Simplify the grammar by removing ε-productions and unreachable symbols.

   $S \rightarrow AB \mid AC$

   $A \rightarrow aAb \mid \varepsilon$

   $B \rightarrow bA$

   $C \rightarrow bCa$

   $D \rightarrow AB$

10. Define context free Grammar for

    i) $L=\{0^i 1^j 2^k \mid j > i+k\}$

    ii) $L=\{w \mid w \in \{a,b\}^*, n_a(w) \neq n_b(w)\}$

Derivations and Parse Trees

(06 Marks)

1. Consider the following grammar:

   $S \rightarrow aAS \mid a$

   $A \rightarrow SbA \mid SS \mid ba$

Give leftmost and rightmost derivations and the parse tree for the following strings:

i) aabbaa

ii) aaaa.

2. Consider the following grammar:

   S → ABC

   A → aA | ε

   B → bB | ε

   C → cC | ε

   Give the leftmost derivation, rightmost derivation and the parse tree for the string aabbba.

3. Obtain the string aaabbabbba by applying left most derivation and the parse tree for the grammar given below. Is it possible to obtain the same string again by applying leftmost derivation but by selecting different productions?

   S→aB|bA

   A→aS|bAA|a

   B→bS|aBB|b

4. Find leftmost and rightmost derivations, and a parse tree for the string +-xyxy using the grammar:

   E → +EE | ∗EE | -EE | x | y

5. Consider the grammar E → E+E | E*E / E-E / x / y. Find leftmost and rightmost derivation and construct corresponding parse trees for the string x+y*x-y.

6. Consider the grammar E → E+E | E*E / E-E / x / y. Find leftmost and rightmost derivation and construct corresponding parse trees for the string x+y*x-y.

7. Obtain the Leftmost and rightmost derivation and parse tree for the string aaabbabbba using the following grammar:

   S→aB|bA

   A→aS|bAA|a

$B \rightarrow bS|aBB|b$

Ambiguity and Elimination of Ambiguity

(05 Marks)

1. Define ambiguous grammar. Verify whether the grammar

   $S \rightarrow aB \mid bA$

   $A \rightarrow aS \mid bAA \mid a$

   $B \rightarrow bS \mid aBB \mid b$

   is ambiguous.

2. What is an ambiguous grammar. Is the grammar

   $S \rightarrow a \mid Sa \mid bSS \mid SSb \mid SbS$

   ambiguous? Justify your answer.

3. What is an ambiguous grammar. Is the grammar

   $S \rightarrow a \mid Sa \mid bSS \mid SSb \mid SbS$

   ambiguous? Justify your answer.

4. Consider the grammar $S \rightarrow SbS \mid a$

This grammar is ambiguous; show in particular that the string abababa has two

i. Parse trees

ii. Leftmost derivations

iii. Rightmost derivations.

5. Show that the grammar

$S \rightarrow SS \mid (S) \mid ()$

is ambiguous.

(06 Marks)

1. Show that the following grammar is ambiguous:

S → aB | bA

A → aS | bAA | a

B → bS | aBB | b

2. Is the grammar below ambiguous? Justify your answer. If the grammar is ambiguous, find an equivalent unambiguous grammar.

   S→ABA

   A→aA | ε

   B→bB | ε

3. Prove that the following grammar is ambiguous:

   S→AB | aaB

   A→a | Aa

   B→b

(07 Marks)

1. Define ambiguous grammar. Show the following grammar is Ambiguous

   S→S+S | S-S | S*S / (S) | S/afor the string a+(a*a)/a-a.

(10 Marks)

1. Define Ambiguity. Consider the grammar

   E → E+E | E*E / (E) | id(i) Find the leftmost, rightmost derivations and parse trees for the string id+id*id(ii) Show that this grammar is ambiguous.

Elimination of Left Recursion

(06 Marks)

1. Consider the given grammar. Eliminate left recursion from the grammars given.

   i)

   S → aAbA | aAbc | ScA

A → aAbab | b

ii)

A → Bxy | x

B → AD

C → A | c

D → d

Left Factoring

(08 Marks)

1. Consider the grammar given below. If not suitable for Predictive Parsing make necessary changes and construct predictive parsing table for the grammar given below. Check whether the grammar is LL(1).

   G:

   A→Bd | a

   B→Ae | b | ε

   C→B | e

Syntax Analysis – Parser, FIRST/FOLLOW, LL(1), Predictive Parsing

FIRST and FOLLOW

(09 Marks)

1. Consider the following Context Free Grammar

   S→SA | 0 | ε

   A→aS1 | a

i. Compute the FIRST and FOLLOW sets for each non-terminal symbol.

ii. Construct the parsing Table for a predictive parser for the grammar.

iii. Is the Grammar LL(1)? Justify.

(10 Marks)

1. Consider the following Context Free Grammar:

   $S \rightarrow SA \mid 0 \mid \varepsilon$

   $A \rightarrow aS1 \mid a$

i. Compute the FIRST and FOLLOW sets for each non-terminal symbol.

ii. Construct the parsing table for a non-recursive predictive parser for the above grammar.

iii. Is the grammar LL(1)? Justify.

Predictive Parsing / Non-Recursive Predictive Parsing

(08 Marks)

1. Consider the grammar given below. If not suitable for Predictive Parsing make necessary changes and construct predictive parsing table for the grammar given below. Check whether the grammar is LL(1).

   G:

   $A \rightarrow Bd \mid a$

   $B \rightarrow Ae \mid b \mid \varepsilon$

   $C \rightarrow B \mid e$

LR Parsing / SLR / LR(0)

(09 Marks)

1. Construct LR(0) parse table for the Grammar Given

   G: $S \rightarrow L$

   $L \rightarrow (L)L \mid a \mid \varepsilon$

Do parsing for the input: (a)$

2 . Given the following grammar, construct LR(0) parse table

CFG:

$S \rightarrow nVn \mid pVp$

$V \rightarrow c$

1. Consider the grammar

   G:

   $S \rightarrow xABz$

   $A \rightarrow Ayw$

   $A \rightarrow y$

   $B \rightarrow u$

i) Construct the LR(0) set of items by indicating the Kernel and Non-kernel items for each item set.

ii) Construct LR(0) parse table.

iii) Show the actions of a parser on input xyuz$

3. Consider the grammar

   G:-

   $S \rightarrow aSb \mid ab$

i) Construct the LR(0) set of items by indicating the Kernel and Non-kernel items for each item set.

ii) Construct LR(0) parse table.

iii) Show the actions of a parser on input aabb$

4. Consider the grammar given below.

   G:

   $S \rightarrow aTRe$

   $T \rightarrow Tbc$

   $T \rightarrow b$

R→d

i) Construct the LR(0) set of items by indicating the Kernel and Non-kernel items for each item set.

ii) Construct LR(0) parse table.

iii) Show the actions of a parser on input abde$

(10 Marks)

1. Construct LR(0) parse table for the Grammar Given

   G: S→L

   L→(L)L | a | ε

Do parsing for the input: (a)$

1. Consider the grammar

   G:-

   S→aSb | ab

i) Construct the LR(0) set of items by indicating the Kernel and Non-kernel items for each item set.

ii) Construct LR(0) parse table.

iii) Show the actions of a parser on input aabb$

Shift Reduce Parsing / Handles

(06 Marks)

1. Illustrate how the handles are identified for the given grammar by constructing a shift reduce parser. Identify the handles for the given inputs string “abbcde” by showing the moves made by a shift reduce parser.

   G:-

   S → aABe

   A → Abc | b

B → d

2. Consider the following grammar:

   S → SS+ | SS∗ | a

   Input string : aaa*a++$

Indicate the configurations of a shift-reduce parser on the above input. In the case of a reduce action, indicate which production is used. With each action, indicate whether there is conflict or not, if exists specify the type of conflict.

(08 Marks)

1. Illustrate how the parser stack is implemented for bottom up strategy with the help of the given desk calculator grammar

   E→E+T | T

   T→T∗F | F

   F→num

Canonical LR(1) Parsing

(06 Marks)

1. Prove that the given Context Free Grammar is LR(1) by generating the parse table.

   S→AaAb | BbBa

   A→ε

   B→ε

(07 Marks)

1. Prove that the given Context Free Grammar is LR(1) by generating the parse table.

   S→AaAb | BbBa

   A→ε

$B \rightarrow \varepsilon$

Canonical LR(1) Items / Canonical LR(1) Parse Table

(10 Marks)

1. Prove that the given Context Free Grammar is LALR(1) by generating the parse table.

   $S \rightarrow Aa \mid bAc \mid dc \mid bda$

   $A \rightarrow d$

LALR Parsing Tables

(10 Marks)

1. Prove that the given Context Free Grammar is LALR(1) by generating the parse table.

   $S \rightarrow Aa \mid bAc \mid dc \mid bda$

   $A \rightarrow d$

2. Consider the following grammar
   $S \rightarrow L=R \mid R$
   $L \rightarrow *R \mid id$
   $R \rightarrow L$

The LR(1) item sets are given below.

Identify LR(1) item sets that can be merged to construct LALR(1) item sets. Mention which sets were merged and why.

Augmented Grammar

(05 Marks)

1. Explain the need of Augmentation in LR grammars and how the given grammar can be changed to an augmented grammar.

Parser Basics / Role of Parser / Error Recovery

No direct full descriptive questions found from the provided images specifically on:

- Role of Parser
- Top-down Parsing theory
- Error Recovery in Predictive Parsing

(Only application-based questions were present.)

IDK—-

Chomsky Normal Form / Simplification of CFG

(10 Marks)

1. Begin with the grammar:
   S → ABC | BaB
   A → aA | BaC | aaa
   B → bBb | a | D
   C → CA | AC
   D → ε

i. Eliminate ε-productions.

ii. Eliminate any unit productions in the resulting grammar.

iii. Eliminate any useless symbols in the resulting grammar.

iv. Put the resulting grammar into Chomsky Normal Form.

(07 Marks)

1. Simplify the following grammar and put it in Chomsky Normal form.
   $S \rightarrow abAB$
   $A \rightarrow bAB \mid \varepsilon$
   $B \rightarrow Baa \mid A \mid \varepsilon$